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Publication date:
2024

Document Version:
Publisher's PDF, also known as Version of record

Link to publication

Citation for pulished version (APA):
Herriott, R. (2024). *Explaining the nature of design*. Paper presented at 18th International Congress on Design Principles and Practices.

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Explaining the Nature of Design: A Pragmatic Definition

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Denmarl

“Things to know and ways of knowing them”

- *Designerly ways of knowing* is a term used in the much-cited paper of the same name (Cross, 1982), but it was first used in *Design in General Education (RCA, 1978)*. Because the 1978 text is almost never cited I am focusing on the 1982 instance of the concept. The key sentence is*: “Design has its own distinct ‘things to know, ways of knowing them, and ways of finding out about them’”* (Cross, 1982, p 221). In particular, the words “*ways of knowing about them*” are under scrutiny.

Distinguishing design from other areas

- Cross was especially concerned with the role of design in education, calling it a third area, in addition to science and the humanities. I will discuss is how design was contrasted by Cross (ibid.) with the sciences and the humanities in exclusive terms. Cross (ibid.) argued that the aim of education in design was not instrumental but for its own sake, for reasons due to design´s intrinsic values. Cross argued that those intrinsic values depended on *“designerly ways of knowing”* (ibid.).

“The central concern of design”

- *“The central concern of Design is ‘the conception and realisation of new things’.*
- *It encompasses the appreciation of ‘material culture’ and the application of ‘the arts of planning, inventing, making and doing’.*
- *At its core is the ‘language’ of ‘modelling’; it is possible to develop students’ aptitudes in this ‘language’, equivalent to aptitudes in the ‘language’ of the sciences - numeracy - and the ‘language’ of humanities - literacy.*
- *Design has its own distinct ‘things to know, ways of knowing them, and ways of finding out about them’.*

The phenomena of study in each culture are

- In the sciences: the natural world.
- In the humanities: human experience.
- In design: the man-made world.

Design is therefore not anything to do with materials (of which the natural world is composed).

Design is therefore not related to anything with human experience.

The appropriate methods in each culture are:

- in the sciences: controlled experiment, classification, analysis
- in the humanities: analogy, metaphor, criticism, evaluation
- in design: modelling, pattern-formation, synthesis.

The methods of design are nothing to do with experiment, classification or analysis,

The methods of design are not related to analogy, metaphor, criticism or evaluation.

The values of each culture are

- in the sciences: objectivity, rationality, neutrality, and a concern for ‘truth’
- in the humanities: subjectivity, imagination, commitment, and a concern for ‘justice’
- in design: practicality, ingenuity, empathy, and a concern for ‘appropriateness.

Designers therefore are not supposed to be objective, neutral or concerned for truth.

Designers are not subjective, imaginative, committed to justice.

What design is and is not

- Cross´s position would be that only *one* item from each category 1-3 could apply at one time. That is to say Cross´s conception implies that to design a car involved

- nothing from the “what” category 1) the natural world and human experience; and
- nothing from the “how” category 2) any of the methods of science and the humanities and
- nothing from the “why” category 3) the values of science and the humanities.

- This leaves, one might say, the rich field of Design somewhat impoverished. It´s interesting this proposition has stood unexamined for so long.

Tidiness and ontology

- “An ontology is an explicit specification of a conceptualization” (Gruber 1994 in Hoekstra, 2009, p 67). What is evident here is that there is an ontological problem rooted in the conception how design is distinguished from science and the humanities.

- There is a human endeavor called design. It would be desirable to support that claim because it is in accord with the first-hand experience of many objective observers. However, I would like to do that while adhering to ontological tidiness and also avoiding the claim that there is a designerly way of knowing (a claim to a special epistemological avenue), because as I will argue there is not.

Articulating

- "*But we are certainly faced with the problem of being more articulate about what it means to be 'designerly' rather than to be 'scientific' or 'artistic'*" (Cross 1982, p 222).

Choices, choices

Do we

- lump design into some other existing group and risk losing sight of some important quality in design (e.g. saying design is a branch of science).

or

- split design off in the manner suggested by Cross and risk generating a precedent that allows for a myriad of ways of knowing (tending to the ontological worst case scenario e.g. everyone has a special way of knowing).

Assertion

There is a finite number of ways (means) of knowing common to all possible observers and these are our sense apparatus and cognitive systems;

The way in which knowledge is accessed does not change its character (though the facts determined may not be exactly same).

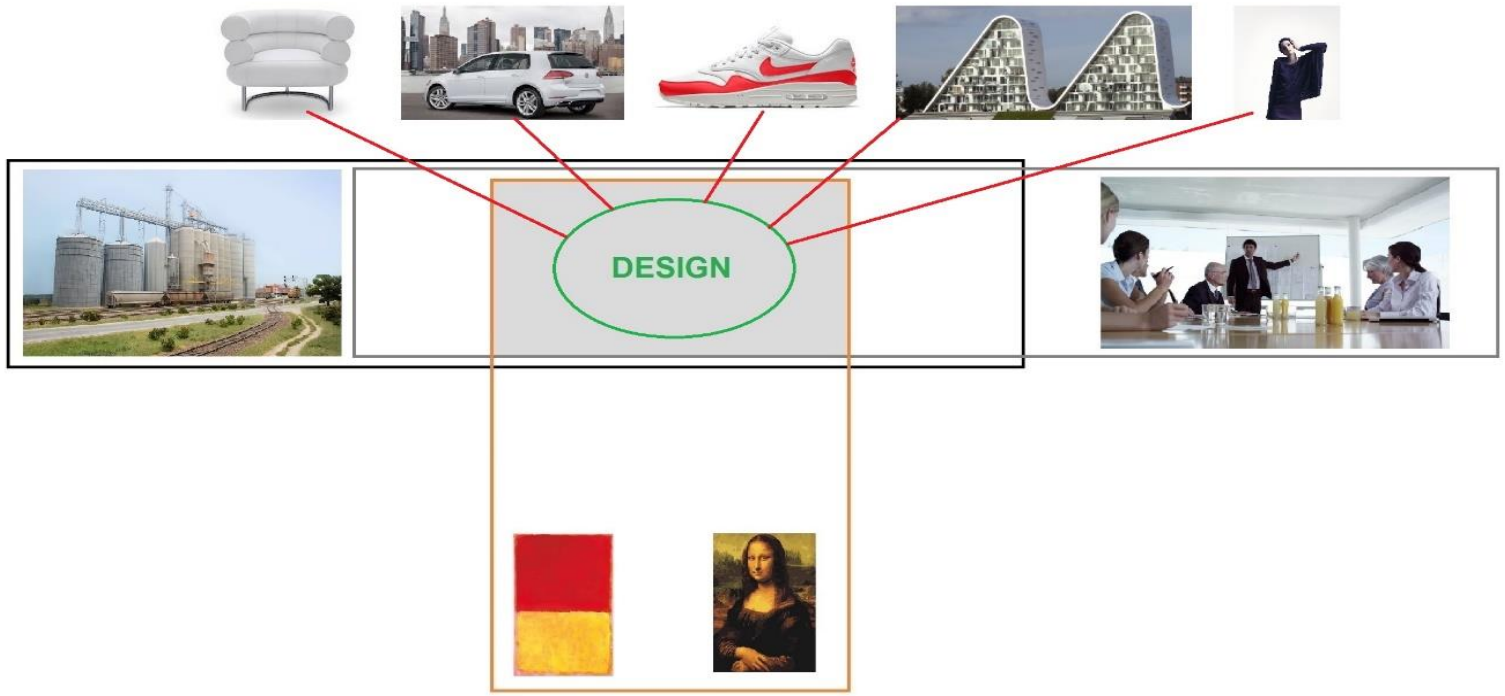
For more on this see:

Herriott, R. (2023). Delightful Phrase: Are there really designerly ways of knowing?. *Design Issues*, *39*(3).

Aside: knowledge is not in artefacts

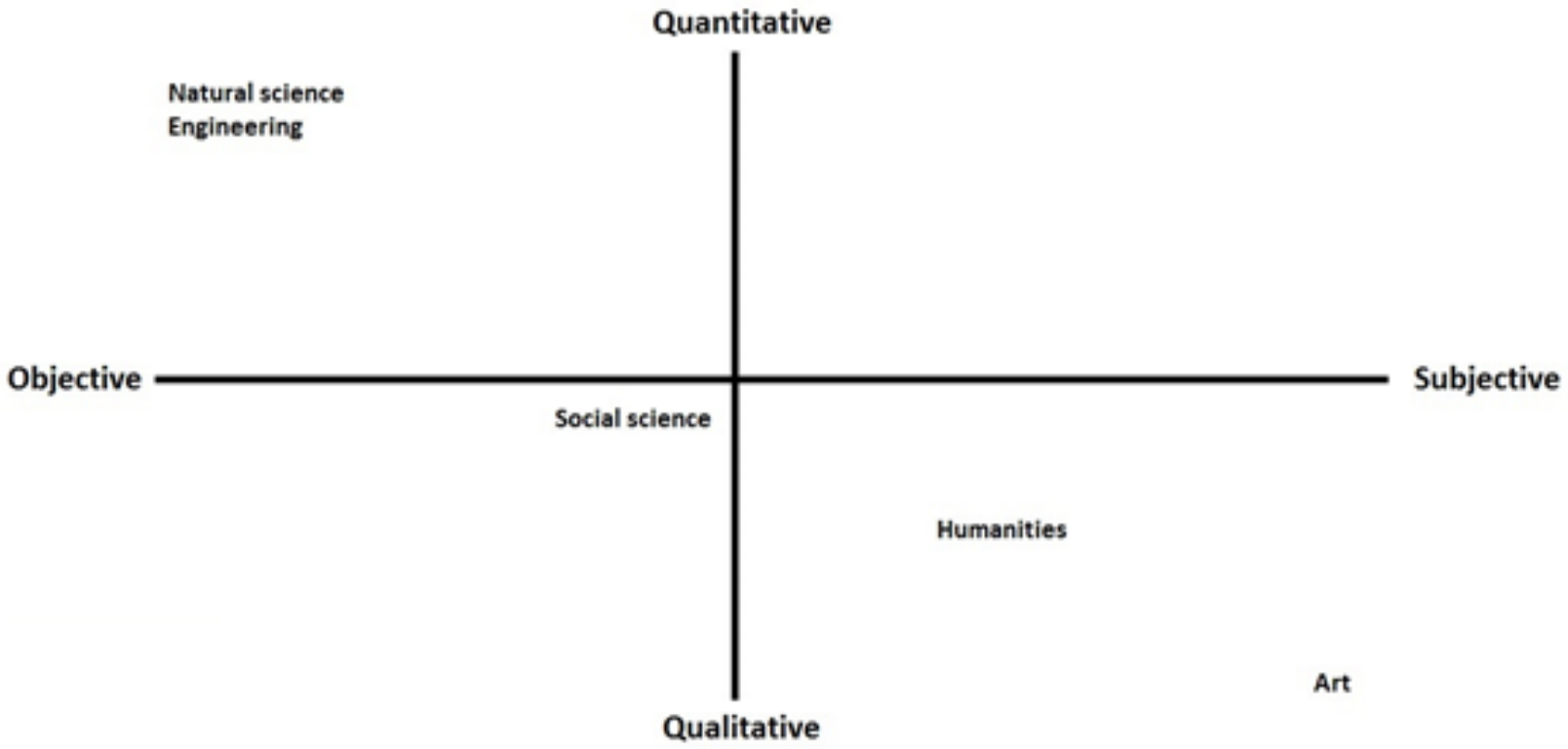


Another way of understanding design

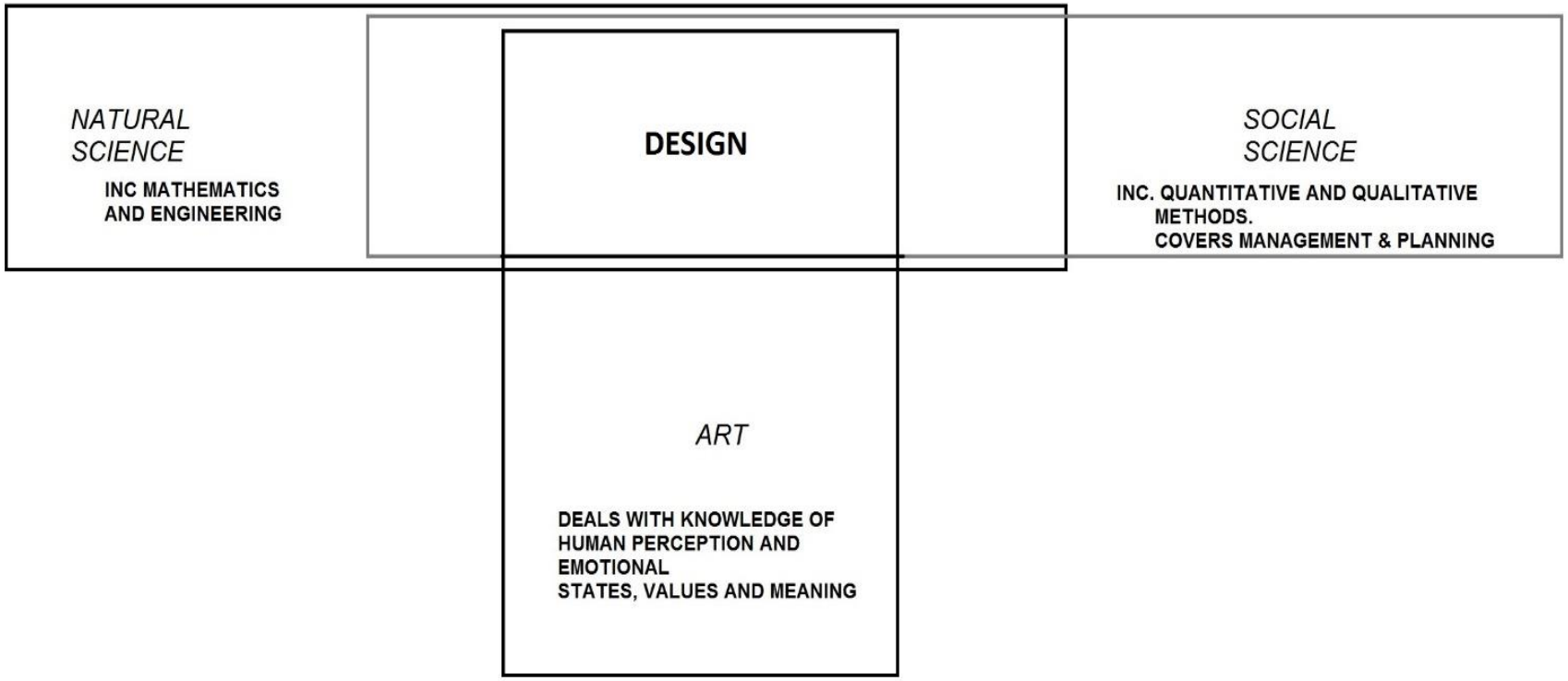


This addresses the manner in which design is at various stages some form of science (attempts to be objective), some form of the humanities (attempts to understand meaning and culture) and some form of art (the subjective, expressive part) while allowing for subsets of other disciplines.

Design: qualitative, quantitative and subjective and objective.



Design: the overlap of fields.



Overlapping domains

- If we apply the overlapping domains model proposed to the car design example we find that:

1) objective, quantitative and qualitative science activities are involved in the engineering and social science research needed to establish a) what the world is like and b) whether the designed proposal works physically and is subjectively acceptable to users;
2) the humanities are present in some part if we imagine the designers are “reading” the culture of the market and the semiotic and semantic content of other designed products (other cars, other objects and also intangible cultural items) and
3) Art methods are used to explore the aesthetic dimension of the product and to create new structures and forms.

Consequences

- This proposal is meaningless unless it has actionable consequences. Does it have any? In this section it is argued that it does. RtD is a mode of research where the researcher does design work to find out about design.

- This argument gives further grounds for design researchers to make greater efforts for making explicit the information gained via design experiments (the designed objects). “Designerly ways of knowing” is not a get-out-of-jail-free card allowing researchers to dodge explicit explanation of their work.

Explicit not implicit

- *"…since RtD is an inquiry process revolving around the making of a product, service, environment, or system, the knowledge gained can be implicit; residing almost entirely within the resulting artifact"* (Zimmermann et al., 2010, p 310).
- Kknowledge does not reside in designed objects.
- Implicit is not an adequate standard – it is not communicable. Some general text with accompanying photos and what are called "architectural interventions" do not attain the standard suggested by Frayling (1993, in Grand and Jonas, 2012, p 185) where the text is there "*to communicate the results, which is what separates research from the gathering of reference materials".*
- **Reliance on implicit understanding of RtD is inadequate. RtD must be constructed to make it explicit and readable**. It is insufficient to carry out design work and gesture at the resultant object with reference to designerly knowledge inherent in objects.

Conclusion

- I argue that design is about methods for creating aesthetically acceptable forms (industrial design) and modes of behaviour (service design).
- Curiously, Cross (1982) does not discuss the one thing that designers do that most other professions don´t, namely to draw a lot.
- The arguments made here lead to an understanding of design that is consistent with the common experience that designers draw things and want things to look as one would draw them. Drawing regains the place it had when design was understood as applied art. Designers are also freed from wondering if the discipline is just a branch of science. They are also freed from defining design in exclusive terms that do not do justice to the fact design synthesizes science, art and the humanities.

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